

Matrix diagonalization

Basic skills

Q1. $M = \begin{pmatrix} 5 & 3 \\ -4 & 2 \end{pmatrix}$ $|M| = 10 + 12 = 22$
 $\text{Adj}(M) = \begin{pmatrix} 2 & -3 \\ 4 & 5 \end{pmatrix}$

so $M^{-1} = \frac{1}{22} \begin{pmatrix} 2 & -3 \\ 4 & 5 \end{pmatrix}$

Q2. For $\text{Adj}(M) = C^T$ where C is the cofactor matrix where a cofactor is the sign alternating determinant of the minor

Q3. a) $\begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$ $|M - \lambda I| = 0$
 $\Rightarrow \begin{vmatrix} -\lambda & 2 \\ 1 & 1-\lambda \end{vmatrix} = 0$

so $\lambda(\lambda-1) - 2 = 0$ $\lambda^2 - \lambda - 2 = 0$

$\Rightarrow (\lambda-2)(\lambda+1) = 0$ $\lambda = -1, 2$

For $\lambda = -1$:

$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} x + 2y = 0 \\ x + 2y = 0 \end{matrix}$

so $y = -\frac{1}{2}x$ so $\begin{pmatrix} x \\ -\frac{1}{2}x \end{pmatrix}$ e.g. $\begin{pmatrix} -1 \\ \frac{1}{2} \end{pmatrix}$ or $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$

For $\lambda = 2$:

$$\begin{pmatrix} -2 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{aligned} -2x + 2y &= 0 \\ x - y &= 0 \end{aligned}$$

so $y = x$ so $\begin{pmatrix} x \\ x \end{pmatrix}$ e.g. $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Q4. $\lambda = 1, 2$ $v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

so $D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ $U = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$

Then $U^{-1} = \frac{1}{3-4} \begin{pmatrix} 3 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}$

so $M = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -3 & 2 \\ 2 & -1 \end{pmatrix}$

Q5. $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\begin{vmatrix} 1-\lambda & 0 \\ 0 & 1-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)^2 = 0$

so $\lambda = 1$ Then $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

so any x, y eigenvectors $\begin{pmatrix} x \\ y \end{pmatrix}$ have
2 degrees of freedom

Q6. a) $M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$ has $\lambda = -1, 2$
 $\rightarrow v = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 see Q3.

so $D = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} U = \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix}$

Then $U^{-1} = \frac{1}{-2-1} \begin{pmatrix} 1 & -1 \\ -1 & -2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix}$

so $M = \frac{1}{3} \begin{pmatrix} -2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix}$

b) $M = \begin{pmatrix} 7 & -3 \\ -2 & 6 \end{pmatrix} \quad |M - \lambda I| = 0$
 $\begin{vmatrix} 7-\lambda & -3 \\ -2 & 6-\lambda \end{vmatrix} = 0$

$\Rightarrow (7-\lambda)(6-\lambda) - 6 = 0 \Rightarrow \lambda^2 - 13\lambda + 36 = 0$

$(\lambda - 9)(\lambda - 4) = 0 \Rightarrow \lambda = 4, 9$

For $\lambda = 4$:

$\begin{pmatrix} 3 & -3 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} 3x - 3y = 0 \\ -2x + 2y = 0 \end{matrix}$

so $y = x$ so $\begin{pmatrix} x \\ x \end{pmatrix}$ e.g. $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

For $\lambda = 9$: $\begin{pmatrix} -2 & -3 \\ -2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} -2x - 3y = 0 \\ -2x - 3y = 0 \end{matrix}$

so $y = -\frac{2}{3}x$ so $\begin{pmatrix} x \\ -\frac{2}{3}x \end{pmatrix}$ e.g. $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$

Then $D = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$ $U = \begin{pmatrix} 1 & 3 \\ 1 & -2 \end{pmatrix}$

so $U^{-1} = \frac{1}{-2-3} \begin{pmatrix} -2 & -3 \\ -1 & 1 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 & 3 \\ 1 & -1 \end{pmatrix}$

so $M = \frac{1}{5} \begin{pmatrix} 1 & 3 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & -1 \end{pmatrix}$

Q7. $M = \begin{pmatrix} 2 & 6 \\ 0 & -1 \end{pmatrix}$ $|M - \lambda I| = 0$

$\begin{vmatrix} 2-\lambda & 6 \\ 0 & -1-\lambda \end{vmatrix} = 0 \Rightarrow -(2-\lambda)(1+\lambda) = 0$

so $\lambda = -1, 2$

For $\lambda = -1$: $\begin{pmatrix} 3 & 6 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} 3x + 6y = 0 \\ \text{so } y = -\frac{1}{2}x \end{matrix}$

so $\begin{pmatrix} x \\ -\frac{1}{2}x \end{pmatrix}$ e.g. $\begin{pmatrix} 1 \\ -\frac{1}{2} \end{pmatrix}$

For $\lambda = 2$: $\begin{pmatrix} 0 & 6 \\ 0 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} 6y = 0, -3y = 0 \\ \text{so } y = 0 \text{ any } x \end{matrix}$

so $\begin{pmatrix} x \\ x \end{pmatrix}$ e.g. $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

then

$$D = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} \quad U = \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix}$$

$$\text{so } U^{-1} = \frac{1}{1+2} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$\text{so } M = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$\text{so } M^4 = (U D U^{-1})^4 = U D U^{-1} U D U^{-1} \dots U D U^{-1} \\ = U D^4 U^{-1}$$

$$= \frac{1}{3} \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} (-1)^4 & 0 \\ 0 & 2^4 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} 1 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 16 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} 1 & 16 \\ -2 & 16 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 33 & 15 \\ 30 & 34 \end{pmatrix}$$

$$= \begin{pmatrix} 11 & 5 \\ 10 & \frac{34}{3} \end{pmatrix}$$

Q8.

$$M = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

$$|M - \lambda I| = 0$$

$$\Rightarrow (4 - \lambda)(3 - \lambda) - 2 = 0$$

$$\Rightarrow \lambda^2 - 7\lambda + 10 = 0 \Rightarrow (\lambda - 5)(\lambda - 2) = 0$$

$$\text{so } \lambda = 2, 5$$

$$\text{For } \lambda = 2: \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y = -x$$

so $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$\text{For } \lambda = 5: \begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y = \frac{1}{2}x$$

so $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$$\text{so } D = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} \quad U = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$$

$$\text{so } U^{-1} = \frac{1}{1+2} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$$

$$\text{so } M = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$$

$$M^n = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & 5^n \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} 2^n & 2 \times 5^n \\ -2^n & 5^n \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$$

$$= \frac{1}{3} \begin{pmatrix} 2^n + 2 \times 5^n & 2 \times 5^n - 2^{n+1} \\ 5^n - 2^n & 2^{n+1} + 5^n \end{pmatrix}$$

Q9. To get a matrix which can't be diagonalised need U to be singular

e.g. $\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$

so need repeated eigenvectors

using $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ $\lambda = 1$ (repeated)

with $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y = 0$

so any x $\begin{pmatrix} x \\ 0 \end{pmatrix}$ e.g. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix}$

so $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ isn't diagonalizable

Q10. $M = \begin{pmatrix} 2 & 8 & -2 \\ -2 & 12 & -2 \\ 4 & -4 & 8 \end{pmatrix}$ has $v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$$\text{SO } \begin{pmatrix} 2 & 8 & -2 \\ -2 & 12 & -2 \\ 4 & -4 & 8 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow 10 = \lambda \quad 10 = \lambda \quad 0 = 0$$

SO $\lambda = 10$ is an eigenvalue

$$|M - \lambda I| = 0 \quad \begin{vmatrix} 2-\lambda & 8 & -2 \\ -2 & 12-\lambda & -2 \\ 4 & -4 & 8-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda) \begin{vmatrix} 12-\lambda & -2 \\ -4 & 8-\lambda \end{vmatrix} - 8 \begin{vmatrix} -2 & -2 \\ 4 & 8-\lambda \end{vmatrix} - 2 \begin{vmatrix} -2 & 12-\lambda \\ 4 & -4 \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda) ((12-\lambda)(8-\lambda) - 8) - 8(2\lambda - 16 + 8) - 2(8 - 48 + 4\lambda) = 0$$

$$\Rightarrow (2-\lambda)(\lambda^2 - 20\lambda + 88) - 16\lambda + 64 + 80 - 8\lambda = 0$$

$$\Rightarrow -\lambda^3 + 22\lambda^2 - 128\lambda + 176 - 24\lambda + 144 = 0$$

$$\Rightarrow \lambda^3 - 22\lambda^2 + 152\lambda - 320 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda^2 - 12\lambda + 32) = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 4)(\lambda - 8) = 0$$

$$\Rightarrow \lambda = 4, 8, 10$$

For $\lambda = 4$:

$$\begin{pmatrix} -2 & 8 & -2 \\ -2 & 8 & -2 \\ 4 & -4 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \underline{0}$$

$$\Rightarrow -2x + 8y - 2z = 0$$

$$4x - 4y + 4z = 0$$

$$\text{so } y = x + z \Rightarrow 6x + 6z = 0$$

$$\text{so } z = -x \Rightarrow y = 0 \text{ so } \begin{pmatrix} x \\ 0 \\ -x \end{pmatrix}$$

$$\text{e.g. } \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

For $\lambda = 8$:

$$\begin{pmatrix} -6 & 8 & -2 \\ -2 & 4 & -2 \\ 4 & -4 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \underline{0}$$

$$\Rightarrow -6x + 8y - 2z = 0, -2x + 4y - 2z = 0,$$

$$4x - 4y = 0 \Rightarrow y = x \Rightarrow 2x - 2z = 0$$

$$\Rightarrow x = z \text{ so } y = x = z \text{ so } \begin{pmatrix} x \\ x \\ x \end{pmatrix}$$

$$\text{e.g. } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\text{so } D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 10 \end{pmatrix} \quad U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{pmatrix}$$

$$\text{so } |U| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -1 & 0 \end{vmatrix} - \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}$$

$$= -1 - (1) + (1) = -1$$

now cofactors of U take minors det

minors

$$\begin{pmatrix} -1 & 1 & 1 \\ -1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}$$

$$\text{so } C = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix} \quad C^T = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 1 & -1 \\ 1 & -2 & 1 \end{pmatrix}$$

$$\text{so } U^{-1} = \frac{1}{-1} \begin{pmatrix} -1 & 1 & 0 \\ -1 & 1 & -1 \\ 1 & -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 1 \\ -1 & 2 & -1 \end{pmatrix}$$

so

$$M = U D U^{-1} =$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 4 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 10 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 1 & -1 & 1 \\ -1 & 2 & -1 \end{pmatrix}$$

$$Q11. M = \begin{pmatrix} a & b-a \\ 0 & b \end{pmatrix}$$

$$|M - \lambda I| = 0 \Rightarrow (\lambda - a)(\lambda - b) = 0$$

$$\Rightarrow \lambda = a, b$$

$$\text{For } \lambda = a: \begin{pmatrix} 0 & b-a \\ 0 & b-a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \underline{0}$$

$$\Rightarrow (b-a)y = 0 \text{ so } y = 0 \text{ so } \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{For } \lambda = b: \begin{pmatrix} a-b & b-a \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \underline{0}$$

$$\Rightarrow (a-b)x + (b-a)y = 0 \Rightarrow y = \frac{b-a}{b-a}x$$

$$\text{so } y = x \text{ so } \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{Then } D = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \quad U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

As $|U| = 0$ not diagonalizable.

so normal method is induction

Assume

$$M^{n-1} = \begin{pmatrix} a^{n-1} & b^{n-1} - a^{n-1} \\ 0 & b^{n-1} \end{pmatrix}$$

$$\text{then } M^n = M M^{n-1} = \begin{pmatrix} a & b-a \\ 0 & b \end{pmatrix} \begin{pmatrix} a^{n-1} & b^{n-1} - a^{n-1} \\ 0 & b^{n-1} \end{pmatrix}$$

$$= \begin{pmatrix} a^n & a(b^{n-1} - a^{n-1}) + b^{n-1}(b-a) \\ 0 & b^n \end{pmatrix}$$

$$= \begin{pmatrix} a^n & b^n - a^n \\ 0 & b^n \end{pmatrix}$$

can show using formal induction

$$\text{Q 12. AS } A \underline{v}_1 = \lambda_1 \underline{v}_1, \quad A \underline{v}_2 = \lambda_2 \underline{v}_2$$

$$B \underline{v}_1 = \mu_1 \underline{v}_1, \quad A \underline{v}_2 = \mu \underline{v}_2$$

$$\Rightarrow M \underline{v}_1 = A(B \underline{v}_1) = A \mu_1 \underline{v}_1$$

$$= \mu_1 A \underline{v}_1 = \mu_1 \lambda_1 \underline{v}_1 = (\lambda_1 \mu_1) \underline{v}_1$$

$$\text{Similarly } M \underline{v}_2 = (\lambda_2 \mu_2) \underline{v}_2$$

SO $M = AB$ has eigenvalues $\lambda_1 \mu_1, \lambda_2 \mu_2$,
eigenvectors $\underline{v}_1, \underline{v}_2$

So $M = U D U^{-1}$

where $U = \begin{pmatrix} \vdots & \vdots \\ \underline{v_1} & \underline{v_2} \\ \vdots & \vdots \end{pmatrix}$ $D = \begin{pmatrix} \lambda_1 \mu_1 & 0 \\ 0 & \lambda_2 \mu_2 \end{pmatrix}$

Q13.

Let $A = U_A D_A U_A^{-1}$

$B = U_B D_B U_B^{-1}$

Then if eigenvectors are equal $U_A = U_B = U$

Then $AB = U D_A U^{-1} U D_B U^{-1} = U D_A D_B U^{-1}$

and $BA = U D_B U^{-1} U D_A U^{-1} = U D_B D_A U^{-1}$

As D_A, D_B diagonal $D_A D_B = D_B D_A$.

So $AB = BA$.

